

INDI Effectiveness Matrices: NED (Indiflight) to FLU (Cybflight)

1 Notation

N_u	Number of actuators (motors)
N_v	Number of pseudo-controls (6: $f_x, f_y, f_z, \tau_\phi, \tau_\theta, \tau_\psi$)
$u_i \in [0, 1]$	Normalised motor command (force-proportional)
$\boldsymbol{\omega}^B$	Body angular rate vector (rad/s)
$\dot{\boldsymbol{\omega}}^B$	Body angular acceleration (rad/s ²)
$\mathbf{a}_{\text{spf}}^B$	Specific force in body frame (m/s ²), measured by accelerometer
$\omega_{\text{mot},i}$	Motor i angular speed (rad/s)
$\dot{\omega}_{\text{mot},i}$	Motor i angular acceleration (rad/s ²)
m	Vehicle mass (kg)
$\mathbf{I}$	Body inertia tensor (kg·m ²), 3×3
T_i	Maximum thrust of motor i (N)
$\mathbf{r}_i = (p_{x,i}, p_{y,i}, 0)$	Motor i position in body frame (m)
s_i	Spin sign: +1 for CW-from-above, −1 for CCW
c_i	Torque coefficient (m): $ \tau_{\text{drag}} = c_i \cdot T_i$

2 Indiflight Definitions (NED / FRD Frame)

Indiflight operates in NED/FRD: x = forward, y = right, z = down. Positive roll = right wing down, positive pitch = nose up, positive yaw = nose right. Thrust acts in $-z$ (upward).

2.1 G1: Static Effectiveness

The G_1 matrix maps motor commands $\mathbf{u}$ to pseudo-controls (specific forces and angular accelerations):

$$\boldsymbol{\nu} = G_1 \mathbf{u}, \quad G_1 \in \mathbb{R}^{6 \times N_u} \quad (1)$$

Column i of G_1 represents the effect of motor i at full throttle ($u_i = 1$):

$$G_1^{\text{NED}}[:, i] = \begin{pmatrix} f_{x,i} \\ f_{y,i} \\ f_{z,i} \\ \alpha_{\phi,i} \\ \alpha_{\theta,i} \\ \alpha_{\psi,i} \end{pmatrix} \quad (2)$$

For standard multirotors with all propellers producing thrust along body $-z_{\text{NED}}$ (upward):

$$f_{x,i} = 0 \quad (3)$$

$$f_{y,i} = 0 \quad (4)$$

$$f_{z,i} = -\frac{T_i}{m} \quad (\text{negative: thrust opposes } +z_{\text{NED}} = \text{down}) \quad (5)$$

The angular acceleration rows are the torque divided by inertia. In NED/FRD, the torque from thrust $\mathbf{F}_i = (0, 0, -T_i)$ at position $\mathbf{r}_i = (p_{x,i}, p_{y,i}, 0)$ is:

$$\boldsymbol{\tau}_i = \mathbf{r}_i \times \mathbf{F}_i + \boldsymbol{\tau}_{\text{drag},i} \quad (6)$$

Computing the cross product ($\mathbf{F}_i = (0, 0, -T_i)$ in NED):

$$\tau_{\phi,i} = p_{y,i} \cdot (-T_i) - 0 = -p_{y,i} T_i \quad (7)$$

$$\tau_{\theta,i} = 0 - p_{x,i} \cdot (-T_i) = +p_{x,i} T_i \quad (8)$$

For the yaw (drag reaction) torque: a CW-from-above propeller has positive ω_z in NED ($z = \text{down}$). The aerodynamic drag opposes the spin, so drag acts on the propeller in $-z_{\text{NED}}$. By Newton's third law the body reaction is also in $-z_{\text{NED}}$ (the air pushes back on the body). Therefore the yaw reaction torque is *negative* for a CW motor ($s_i = +1$):

$$\tau_{\psi,i} = -s_i c_i T_i \quad (9)$$

The angular acceleration effectiveness:

$$\begin{pmatrix} \alpha_{\phi,i} \\ \alpha_{\theta,i} \\ \alpha_{\psi,i} \end{pmatrix} = \mathbf{I}^{-1} \begin{pmatrix} \tau_{\phi,i} \\ \tau_{\theta,i} \\ \tau_{\psi,i} \end{pmatrix} = \mathbf{I}^{-1} \begin{pmatrix} -p_{y,i} T_i \\ +p_{x,i} T_i \\ -s_i c_i T_i \end{pmatrix} \quad (10)$$

For diagonal inertia $\mathbf{I} = \text{diag}(I_{xx}, I_{yy}, I_{zz})$:

$$\alpha_{\phi,i} = -p_{y,i} T_i / I_{xx} \quad (11)$$

$$\alpha_{\theta,i} = +p_{x,i} T_i / I_{yy} \quad (12)$$

$$\alpha_{\psi,i} = -s_i c_i T_i / I_{zz} \quad (13)$$

Units. Force rows: $\text{N/kg} = \text{m/s}^2$. Torque rows: $\text{Nm}/(\text{kg}\cdot\text{m}^2) = \text{rad/s}^2$. Config scaling: `actG1_fz` stored $\times 100$, `actG1_roll/pitch/yaw` stored $\times 10$.

2.2 G2: Rate-Dependent Effectiveness

$G_2 \in \mathbb{R}^{3 \times N_u}$ captures the reaction torque from motor angular acceleration. When motor i accelerates ($\dot{\omega}_{\text{mot},i} \neq 0$), the propeller+rotor inertia $J_{\text{prop},i}$ exerts a reaction torque on the body:

$$\boldsymbol{\tau}_{\text{react},i} = -J_{\text{prop},i} \dot{\omega}_{\text{mot},i} \hat{\mathbf{e}}_z \quad (14)$$

For standard quads, only the yaw axis is affected:

$$G_2^{\text{NED}}[:, i] = \begin{pmatrix} 0 \\ 0 \\ G_{2,\psi,i} \end{pmatrix} \quad (15)$$

The G2 values are configured directly (from system identification) and enter the allocation via the G2 scaler:

$$\text{G2_scaler}_i = \frac{\omega_{\text{max},i}^2}{2\tau_{\text{mot},i}} \quad (16)$$

where $\tau_{\text{mot},i}$ is the motor time constant (s). The combined effectiveness matrix used in the WLS allocation is:

$$\tilde{G}[j, i] = G_1[j, i] + \begin{cases} \frac{\text{G2_scaler}_i}{\omega_{\text{mot},i}} G_2[j - 3, i] & \text{if } j > 2 \\ 0 & \text{if } j \leq 2 \end{cases} \quad (17)$$

2.3 Pseudo-Control (INDI Law)

The incremental pseudo-control error:

$$\Delta \nu = \begin{pmatrix} 0 \\ 0 \\ f_{z,\text{sp}} - f_{z,\text{meas}} \\ \dot{\omega}_{\phi,\text{sp}} - \dot{\omega}_{\phi,\text{meas}} \\ \dot{\omega}_{\theta,\text{sp}} - \dot{\omega}_{\theta,\text{meas}} \\ \dot{\omega}_{\psi,\text{sp}} - \dot{\omega}_{\psi,\text{meas}} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ \sum_i G_2[0, i] \dot{\omega}_{\text{mot},i} \\ \sum_i G_2[1, i] \dot{\omega}_{\text{mot},i} \\ \sum_i G_2[2, i] \dot{\omega}_{\text{mot},i} \end{pmatrix} \quad (18)$$

(Both terms multiplied by `doIndi` flag; when on ground, the controller solves the non-incremental problem instead.)

3 Frame Transform: NED $\rightarrow$ FLU

Cybfight uses FLU: x = forward, y = left, z = up.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}_{\text{FLU}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}_{\text{NED}} \quad (19)$$

Position. x unchanged, y flips, z flips:

$$p_x^{\text{FLU}} = p_x^{\text{NED}}, \quad p_y^{\text{FLU}} = -p_y^{\text{NED}}, \quad p_z^{\text{FLU}} = -p_z^{\text{NED}} \quad (20)$$

Angular rates and torques. Same transform (pseudovectors under proper rotation):

$$\omega_{\phi}^{\text{FLU}} = \omega_{\phi}^{\text{NED}}, \quad \omega_{\theta}^{\text{FLU}} = -\omega_{\theta}^{\text{NED}}, \quad \omega_{\psi}^{\text{FLU}} = -\omega_{\psi}^{\text{NED}} \quad (21)$$

Rotor handedness label vs. yaw torque sign. The rotor handedness label s_i is a geometric property of the propeller (CW-from-above = +1, CCW = -1) and is the same in both frames. However, the *yaw torque* sign changes between frames because the body z -axis flips:

$$\tau_{\psi,i}^{\text{NED}} = -s_i c_i T_i \quad (\text{CW} \rightarrow \text{negative yaw in NED}) \quad (22)$$

$$\tau_{\psi,i}^{\text{FLU}} = +s_i c_i T_i \quad (\text{CW} \rightarrow \text{positive yaw in FLU}) \quad (23)$$

This is consistent: $\tau_{\psi}^{\text{FLU}} = -\tau_{\psi}^{\text{NED}}$ (z -axis flips). Cybfight's `SpinDir::Cw = 1` directly gives the FLU sign.

4 Cybflight G1 Derivation (FLU Frame)

4.1 From First Principles

In FLU, thrust acts in $+z$ (upward). For motor i with $\mathbf{F}_i = (0, 0, +T_i)$ at $\mathbf{r}_i = (p_{x,i}^{\text{FLU}}, p_{y,i}^{\text{FLU}}, 0)$:

$$\boldsymbol{\tau}_i^{\text{FLU}} = \mathbf{r}_i \times \mathbf{F}_i + \boldsymbol{\tau}_{\text{drag},i} \quad (24)$$

Cross product:

$$\tau_{\phi,i}^{\text{FLU}} = p_{y,i}^{\text{FLU}} \cdot T_i \quad (25)$$

$$\tau_{\theta,i}^{\text{FLU}} = -p_{x,i}^{\text{FLU}} \cdot T_i \quad (26)$$

$$\tau_{\psi,i}^{\text{FLU}} = s_i^{\text{FLU}} \cdot c_i \cdot T_i \quad (27)$$

These match `mixer.rs` lines 151–155 exactly.

The INDI G1 in FLU:

$$G_1^{\text{FLU}}[:, i] = \begin{pmatrix} 0 \\ 0 \\ T_i / m \\ \mathbf{I}^{-1} \begin{pmatrix} p_{y,i}^{\text{FLU}} T_i \\ -p_{x,i}^{\text{FLU}} T_i \\ s_i^{\text{FLU}} c_i T_i \end{pmatrix} \end{pmatrix} \quad (28)$$

For diagonal inertia:

$$G_1^{\text{FLU}}[0, i] = 0 \quad (29)$$

$$G_1^{\text{FLU}}[1, i] = 0 \quad (30)$$

$$G_1^{\text{FLU}}[2, i] = +T_i / m \quad (\text{positive} = \text{upward in FLU}) \quad (31)$$

$$G_1^{\text{FLU}}[3, i] = +p_{y,i}^{\text{FLU}} T_i / I_{xx} \quad (\text{roll}) \quad (32)$$

$$G_1^{\text{FLU}}[4, i] = -p_{x,i}^{\text{FLU}} T_i / I_{yy} \quad (\text{pitch}) \quad (33)$$

$$G_1^{\text{FLU}}[5, i] = +s_i^{\text{FLU}} c_i T_i / I_{zz} \quad (\text{yaw}) \quad (34)$$

4.2 From Existing Cybflight G1 (Physical Units)

Cybflight's `mixer.rs` already computes a 4-row G1 in physical units (FLU):

$$G_{1,\text{phys}}^{\text{FLU}}[:, i] = \begin{pmatrix} T_i \\ p_{y,i}^{\text{FLU}} T_i \\ -p_{x,i}^{\text{FLU}} T_i \\ s_i^{\text{FLU}} c_i T_i \end{pmatrix} = \begin{pmatrix} \text{thrust (N)} \\ \text{roll torque (N m)} \\ \text{pitch torque (N m)} \\ \text{yaw torque (N m)} \end{pmatrix} \quad (35)$$

The INDI G1 is obtained by converting to acceleration space:

$$G_1^{\text{FLU}}[:, i] = \begin{pmatrix} 0 \\ 0 \\ G_{1,\text{phys}}[0, i] / m \\ \mathbf{I}^{-1} \begin{pmatrix} G_{1,\text{phys}}[1, i] \\ G_{1,\text{phys}}[2, i] \\ G_{1,\text{phys}}[3, i] \end{pmatrix} \end{pmatrix} \quad (36)$$

This handles non-diagonal inertia correctly via full $\mathbf{I}^{-1}$.

5 Cybflight G2 Derivation (FLU Frame)

G2 follows the same sign convention as the torque rows of G1. For standard multirotors (vertical motors), only yaw is affected:

$$G_2^{\text{FLU}}[:, i] = \begin{pmatrix} 0 \\ 0 \\ G_{2,\psi,i}^{\text{FLU}} \end{pmatrix} \quad (37)$$

The yaw G2 sign equals the G1 yaw sign for the same motor:

- CW motor ($s_i = +1$): speeding up $\Rightarrow$ reaction torque in $+z_{\text{FLU}} \Rightarrow$ positive yaw G2
- CCW motor ($s_i = -1$): speeding up $\Rightarrow$ reaction torque in $-z_{\text{FLU}} \Rightarrow$ negative yaw G2

G2 values come from system identification, not first principles. The combined effectiveness matrix (same as Eq. 17) uses the G2 scaler to adapt based on operating point:

$$\tilde{G}^{\text{FLU}}[j, i] = G_1^{\text{FLU}}[j, i] + \begin{cases} \frac{\text{G2_scaler}_i}{\omega_{\text{mot},i}} G_2^{\text{FLU}}[j - 3, i] & j > 2 \\ 0 & j \leq 2 \end{cases} \quad (38)$$

6 Numerical Verification

Using cybflight's SAKURAH743 quad parameters:

$$m = 0.55 \text{ kg}, \quad I_{xx} = 0.0025, \quad I_{yy} = 0.0021, \quad I_{zz} = 0.0043 \text{ kg m}^2 \\ T_i = 8.5 \text{ N}, \quad c_i = 0.022 \text{ m}$$

	M0 (RR, CW) $p = (-d, -d)$	M1 (FR, CCW) $p = (+d, -d)$	M2 (RL, CCW) $p = (-d, +d)$	M3 (FL, CW) $p = (+d, +d)$
f_z (m/s ²)	+15.45	+15.45	+15.45	+15.45
α_ϕ (rad/s ²)	-340.0	-340.0	+340.0	+340.0
α_θ (rad/s ²)	+303.6	-303.6	+303.6	-303.6
α_ψ (rad/s ²)	+43.5	-43.5	-43.5	+43.5

where $d_x = 0.075$ m, $d_y = 0.1$ m. Physical checks:

- All $f_z > 0$: thrust is upward in FLU. ✓
- M0 (rear-right): $\alpha_\phi < 0$ (right motor $\Rightarrow$ left-side-up roll). ✓
- M0 (rear): $\alpha_\theta > 0$. In FLU, positive τ_y rotates $+x$ toward $-z$ via $\boldsymbol{\omega} \times \mathbf{r}$, i.e. nose down. Rear motor thrusting up pitches nose down. ✓
- M0 (CW): $\alpha_\psi > 0$ (CW reaction = $+z_{\text{FLU}}$). ✓
- Yaw signs: CW motors (M0, M3) positive, CCW (M1, M2) negative. ✓
- Roll: right motors (M0, M1, $p_y < 0$) negative, left (M2, M3) positive. ✓
- Pitch: rear motors (M0, M2, $p_x < 0$) positive, front (M1, M3) negative. ✓

Cross-check with NED. Applying the frame transform to the FLU values:

$$\begin{aligned}
G_1^{\text{NED}}[2, \text{M0}] &= -G_1^{\text{FLU}}[2, \text{M0}] = -15.45 & (-z = \text{upward in NED}) \\
G_1^{\text{NED}}[3, \text{M0}] &= +G_1^{\text{FLU}}[3, \text{M0}] = -340.0 & (\text{roll, } x \text{ unchanged}) \\
G_1^{\text{NED}}[4, \text{M0}] &= -G_1^{\text{FLU}}[4, \text{M0}] = -303.6 & (\text{pitch, } y \text{ flips}) \\
G_1^{\text{NED}}[5, \text{M0}] &= -G_1^{\text{FLU}}[5, \text{M0}] = -43.5 & (\text{yaw, } z \text{ flips})
\end{aligned}$$

In NED: CW motor yaw is negative (drag reaction in $-z_{\text{NED}}$), rear motor pitch is negative (nose-down in NED sign convention). All consistent with indiflight. ✓